



User Guide: Iron-Dome NIDS Implementation

1. Introduction

This manual outlines the complete process to download, configure, and operate the **Iron-Dome Network Intrusion Detection System**. This application leverages Machine Learning (Random Forest) to analyze network traffic patterns and distinguish between benign behavior and malicious attacks (DDoS/Port Scans) in real-time.

Key Capabilities:

- **Hybrid Data Engine:** Train on internal simulation logic or external real-world datasets (CIC-IDS2017).
- **Live Attack Simulator:** Manually inject packet data to test the system's defense.
- **Forensic Reporting:** Generate automated PDF security audits.

2. Prerequisites

Ensure your host machine meets the following requirements before installation:

- **Operating System:** Windows 10/11, macOS, or Linux.
- **Python:** Version 3.10 or higher.



CRITICAL: During installation on Windows, you **MUST** check the box "Add Python to PATH".

- **Version Control:** Git installed (for downloading the project).
- **Code Editor:** Visual Studio Code (Recommended).

3. Installation & Setup

Step 1: Download the Repository

Instead of creating folders manually, you will download the complete structured project from GitHub.

1. Open your **Command Prompt** (Windows) or **Terminal** (Mac/Linux).
2. Navigate to your Desktop: `cd Desktop`
3. Clone the repository:

```
git clone https://github.com/[YOUR_USERNAME]/Iron-Dome-NIDS.git
```

4. Enter the project directory: `cd Iron-Dome-NIDS`

Step 2: Install Dependencies

The project relies on specific Data Science libraries.

1. Open the folder in **VS Code** (code .).
2. Open the VS Code Terminal (**Ctrl + `**).
3. Execute the installation command: `pip install -r requirements.txt`

Note: If you encounter permission errors, try: `pip install --user -r requirements.txt`

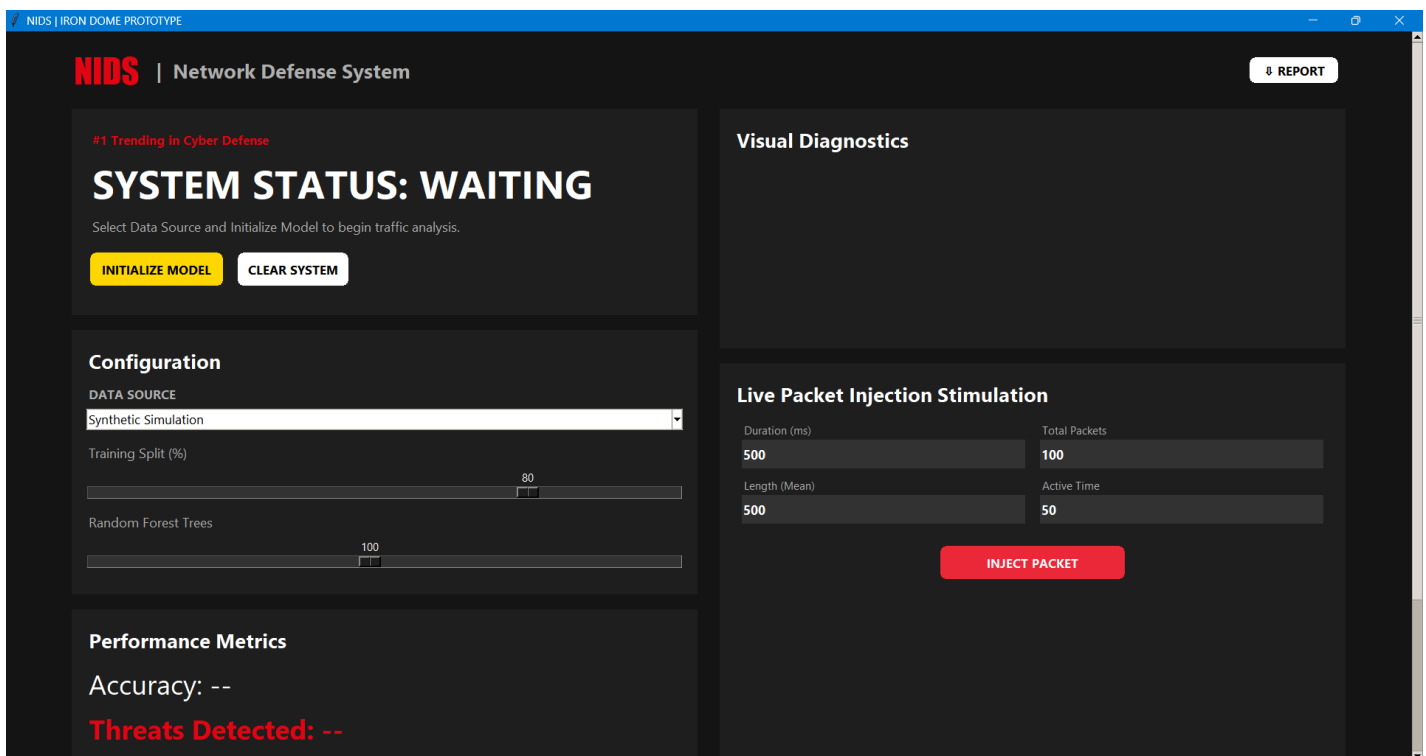
4. Execution

Step 1: Launch the System

In the VS Code Terminal, run the main interface script: `python src/main.py`

Step 2: Verify Startup Status

- A window titled "NIDS | IRON DOME PROTOTYPE" will launch.



- **System Status:** Should display "WAITING".
- **Terminal Log:** The bottom panel should read:

```
SYSTEM LOGS
[13:30:57] root@nids:~$ NIDS KERNEL INITIALIZED...
```

5. Operational Walkthrough

Mode A: Synthetic Simulation (The Quick Demo)

Use this mode for quick presentations without needing external files.

1. **Configuration:** Under "DATA SOURCE", ensure "**Synthetic Simulation**" is selected.

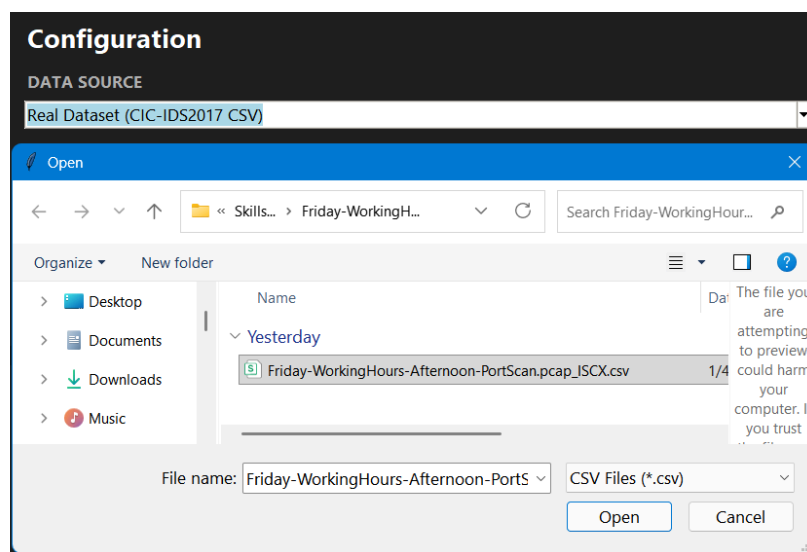


2. **Initialize:** Click the **Yellow "INITIALIZE MODEL"** button.
3. **Observation:**
 - The system generates 5,000 mock traffic records.
 - The AI trains in real-time.
 - **Performance Metrics** (Bottom Left) will update (e.g., Accuracy: 99.9%).
 - **Visual Diagnostics** (Right) will render the Confusion Matrix.

Mode B: Real-World Data Analysis (Advanced)

Use this mode to validate against the CIC-IDS2017 Benchmark.

1. **Download Data:** Obtain the [Friday-WorkingHours...pcap_ISCX.csv](#) from [Kaggle](#).
2. **Configuration:** Change the "DATA SOURCE" dropdown to **"Real Dataset (CIC-IDS2017 CSV)"**.
3. **Load File:** A file explorer window will automatically pop up. Select your downloaded [.csv](#) file.



4. **Initialize:** Click the **Yellow "INITIALIZE MODEL"** button. The system will now train on actual historical attack data.

Mode C: Live Packet Injection (Attack Simulation)

Test the AI's reaction to different traffic patterns.

Scenario 1: The Normal User (Benign)

- **Input Values:**
 - Duration: [5000](#) (ms)
 - Total Packets: [5](#)

- Length Mean: 500
- Active Time: 50
- **Action:** Click the **Red "INJECT PACKET"** button.
- **Result:** Log displays "Packet verified. Traffic benign." (No Alarm).

Scenario 2: The DDoS Attack (Malicious)

- **Input Values:**
 - Duration: 10 (ms) *Suspiciously fast*
 - Total Packets: 300 *Suspiciously high volume*
 - Length Mean: 10
 - Active Time: 10
- **Action:** Click the **Red "INJECT PACKET"** button.
- **Result:**
 - **Popup Alert:** "NIDS ALERT: High-severity threat detected!"
 - **Log:** "ALERT: MALICIOUS PACKET DROPPED."

Mode D: Reporting & Reset

1. **Generate Audit:** Click the **White "REPORT"** button (Top Right). Save the PDF. Open it to see a professional summary and a guide on reading the graphs.
2. **Reset:** To start over, click the **White "CLEAR SYSTEM"** button.

6. Troubleshooting

Issue	Cause	Solution
<code>ModuleNotFoundError</code>	Libraries not installed.	Run <code>pip install -r requirements.txt</code> again.
<code>No module named 'src'</code>	Running from the wrong folder.	Ensure you are in the root Iron-Dome-NIDS folder NOT inside <code>src/</code> .
CSV File Not Loading	File format mismatch.	Ensure you downloaded the correct "MachineLearningCSV" version from Kaggle.
UI Text is Blurry	Windows Scaling.	The code includes a High-DPI fix, but ensure your monitor scale is 100% or 125%.

7. Support & References

- **Project Repository:** <https://github.com/hi-ashup/VOIS--Iron-Dome-NIDS>
- **Dataset Source:** [Canadian Institute for Cybersecurity \(CIC-IDS2017\)](#).
- **License:** GNU General Public License v2.0 (GPLv2).